



Temperature-dependent conductivity in room temperature ionic liquids. The relative importance of diffusional and exchange mechanisms

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Abstract

We report on the translational diffusion behavior of the chromophore Rhodamine B (RB) in a series of alkylimidazolium room temperature ionic liquids (RTILs), 1-ethyl-3-methylimidazolium bis(trifluoromethylsulfonyl) imide (EMIM TFSI), 1-*n*-butyl-3-methylimidazolium bis(trifluoromethylsulfonyl) imide (BMIM TFSI), 1-methyl-3-*n*-octylimidazolium bis(trifluoromethylsulfonyl) imide (OMIM TFSI), and 1-*n*-butyl-3-methylimidazolium tetrafluoroborate (BMIM BF₄). We compare these results to published data on the electrical conductivity of the same RTILs. This comparison underscores the inability of conductivity in RTILs to be accounted for solely on the basis of constituent ion translational diffusion as described in the Nernst-Einstein model. The dependence of the “excess” conductivity on RTIL cation structure and system temperature points to the important role of exchange-mediated ion transport in RTILs, and from the temperature-dependence of the excess conductivity, we determine the activation energy for the exchange process. The activation energy for the exchange process is less than a typical hydrogen bond, suggesting a facile process, and the cation structure dependence of the activation energy is consistent with the intrinsically heterogeneous nature of RTILs.

Introduction

Room temperature ionic liquids (RTILs) are a remarkably versatile class of materials, finding application in areas ranging from organic synthesis to carbon capture, gas sequestration, drug delivery, and space propulsion [1–12]. Among the favorable materials properties of RTILs are the characteristically broad range of temperatures over which they exist in the liquid phase, their extremely low vapor pressure, and wide electrochemical window. Despite extensive investigation of RTILs and the large number of applications in which they are used, achieving a fundamental understanding of the molecular-scale and mesoscopic properties of RTILs remains to be achieved. We are particularly interested in the operative mechanism(s) of constituent mobility and conductivity in RTILs because of their central role in devices such as electric double layer capacitors and

redox flow batteries [13–21]. RTILs are known to be heterogeneous media and to exhibit organization over a range of length scales [22–33], and the dynamics of ion and ion-pair diffusion and exchange continue to be investigated [34, 35]. The basis for the ongoing challenges in understanding RTIL molecular-scale processes lie in the high charge density, rapid exchange dynamics and strong molecular-scale interactions. For example, there is a rich literature dedicated to understanding the extent to which RTILs exist as dissociated ions, with estimates ranging from almost no dissociation to almost complete dissociation. Following decades of effort, it is now understood that most RTILs are approximately 50% dissociated [36].

Other application areas where RTILs have been utilized include energy storage, release, and generation via the piezoelectric effect [37–40]. In these application areas, the electrical and ionic conductivity of RTILs is central to their utility, and because of the materials properties of RTILs, gaining a predictive understanding of their conductivity has proven to be a challenge. Many treatments of conductivity in RTILs find that the Nernst-Einstein equation does not provide sufficient accuracy [34, 41, 42]. This is not surprising because the Nernst-Einstein equation is intended to describe

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the behavior of dilute solutions of ions in molecular solvents such as water. The Nernst-Einstein equation does serve as a useful starting point, however, and the central question is what mechanism(s) can account for conductivity in excess of the predictions of this model. It is the purpose of this work to address that question. We have evaluated the diffusional properties of RTIL constituent ions as a function of temperature using fluorescence recovery after photobleaching (FRAP) measurements of the fluorophore Rhodamine B (RB) in four imidazolium cation-containing ionic liquids. Our experimental data, when compared to an abundance of electric conductivity data reported in the literature [41–52], point to the central role of ion exchange dynamics in explaining the conductivity of RTILs [53, 54]. From the temperature-dependence of our results, we can determine the activation energy of ion exchange and relate it to RTIL cation structure and local organization. These results are a consequence of nanoscale heterogeneity in imidazolium RTILs and the role of this heterogeneity in mediating charged species mobility.

Experimental Methods

Materials

The ionic liquids 1-ethyl-3-methylimidazolium bis(trifluoromethylsulfonyl) imide (EMIM TFSI) (> 98%, TCI), 1-*n*-butyl-3-methylimidazolium bis (trifluoromethylsulfonyl) imide (BMIM TFSI), (> 99%, TCI), 1-methyl-3-*n*-octylimidazolium bis(trifluoromethylsulfonyl) imide (OMIM TFSI) (> 97%, TCI), and 1-*n*-butyl-3-methylimidazolium tetrafluoroborate (BMIM BF₄) (≥ 98%, Sigma Aldrich) were purified before use by a procedure described elsewhere [34]. The chromophore Rhodamine B (RB) (97%, Sigma Aldrich) was used as received for FRAP studies. Structures of each of the cations and anions contained in these ionic liquids, and the structure of RB, are shown in Fig. 1. To remove water and other impurities, the

ionic liquids were stored over activated carbon in a nitrogen purged glove bag before syringe filtration (0.22 μm, Durapore, Millex). All glassware used in this purification procedure was dried at 200 °C prior to use. The filtered ionic liquid was heated to 85 °C in a Schlenk flask for at least five hours while being sparged with argon. Karl Fischer titration (Mettler Toledo, C10SD) of each ionic liquid was used to determine water content of less than 50 ppm following purification. Solvents ethanol (anhydrous, ≥ 99.5%, Sigma-Aldrich) and 2-propanol (≥ 99.5%, Sigma-Aldrich) were used as received, without further purification.

Sample preparation

Each ionic liquid was prepared in a solution with the chromophore RB using a stock chromophore solution of ca. 5×10^{-4} M in ethanol. The procedure for the ionic liquid solutions made to a final RB concentration of ca. 5×10^{-6} M has been described previously [34]. Briefly, an aliquot of RB stock solution was placed in a vessel and evaporated to dryness with a stream of N₂(g), followed by addition of the RTIL, and stirred to take up the chromophore into the RTIL. Materials in the sample cell were all cleaned prior to assembly. Coverslips (Globe Scientific Inc., #1.5) were cleaned in Piranha solution (*Caution! Strong oxidizer!*) and rinsed in Milli-Q water (Millipore, 18 MΩ-cm) before rinsing with ethanol and drying with a stream of N₂(g). The glass microscope slides were sonicated for fifteen minutes in detergent (Fisher Sparkleen 1[®]) solution made with DI water then sonicated in Milli-Q water for fifteen minutes and finally sonicated in 2-propanol for fifteen minutes. Prior to assembly, the slides were rinsed with ethanol and dried with a stream of N₂(g). Adhesive spacers that contained well cutouts (0.12 mm depth, SecureSeal, Sigma Aldrich) were adhered to the microscope slide and approximately 5 μL of a chromophore-containing RTIL solution was deposited in the well while contained in a N₂-purged glove bag. The coverslip was subsequently adhered to the top of the spacer, sealing the RTIL solution between the slide and spacer.

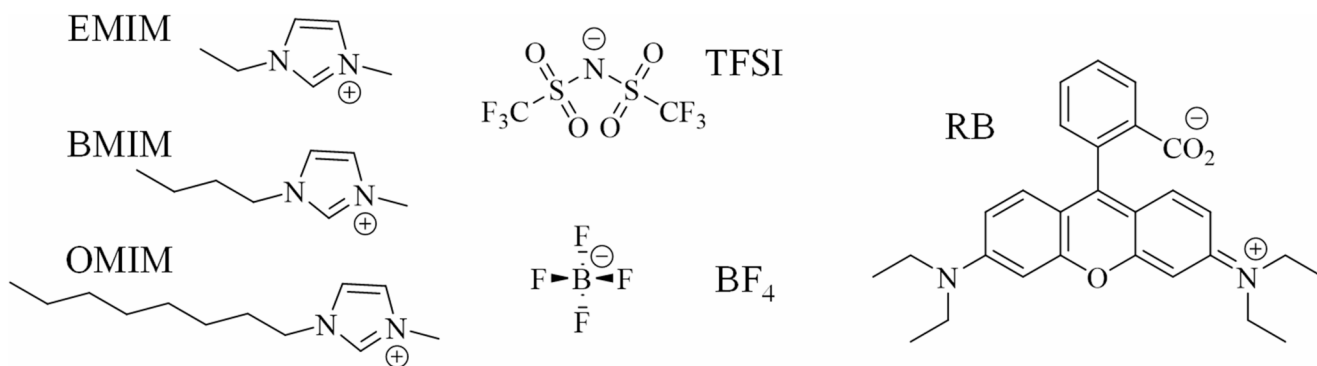


Fig. 1 Structures of the ionic liquid cations and anions, and chromophore used in this work

Sample measurements

Fluorescence Recovery After Photobleaching (FRAP) measurements were performed using an inverted confocal microscope (Nikon Eclipse TiU), a laser source (Nikon LU-N4) operated at 560 nm, and a three-color imaging detector (Nikon C2). A temperature-controlled microscope stage system consisting of a benchtop water circulator (INTEC, C100W-P1), a precision temperature controller (INTEC, mK2000B), and a thermal TEC stage (INTEC, TS10SI) was placed on the inverted microscope stage. Temperature readouts and stage temperature control were monitored using commercial software from the vendor (INTEC IntecApp-ProcessControl, v. 11.00.02). FRAP control and acquisition software (Nikon Elements, 4.30.02) was used to acquire data. MATLAB code written in-house was used to fit the FRAP recovery curves. A 20 $\times$ microscope objective was used for all measurements, and more than ten samples of each chromophore-containing RTIL solution were acquired at room temperature (21 °C), then at 10 °C, 20 °C, 30 °C, 40 °C, 50 °C, and 55 °C using the temperature-controlled stage. The stage temperature was held constant at the temperature of interest for the duration of each measurement. The next measurement commenced only when the selected stage temperature had stabilized. An example of the stage temperature readout during the course of a series of measurements for a given RTIL is shown in Fig. S1.

Results and discussion

As noted above, several of the important application areas in which RTILs are utilized rely on the conductivity of the RTIL and the mobilities of its constituent ions. For example, the pressure-induced piezoelectric response of RTILs requires that the transient crystalline species formed upon the application of pressure maintain electrical contact with the electrodes in the cell for the piezoelectric response to be detected [37–39]. For the use of RTILs in electric double-layer capacitors, conductivity and charge mobility are essential properties for device operation. The conductivity of RTILs has been investigated extensively, for both fundamental and applied reasons, and it is known that conductivity in this family of materials cannot be accounted for fully by diffusive motion of the dissociated (ionic) constituents. We have undertaken a series of studies focusing on the processes contributing to the conductivity of RTILs because of the potential for such information to facilitate a deeper understanding of molecular- and nano-scale dynamics and heterogeneity within these systems and the implications for the utility of these materials in a host of electrochemical and energy-related applications.

We previously investigated the rotational and translational diffusion dynamics of fluorescent chromophores dissolved in BMIM TFSI and found that diffusional motion alone could not account for the conductivity observed experimentally [34]. There was an additional contribution to conductivity in this RTIL, which we ascribed to charge migration mediated by the exchange dynamics of the anionic and cationic RTIL constituents arising from association and dissociation. Exchange dynamics can give rise to charge transport in the medium that does not require large-scale translational motion of the ions [53, 54]. Such a process is expected to depend sensitively on the proximity of the ionic (dissociated) RTIL constituents, and on any local organization in the RTIL. Subsequent FRAP experiments on BMIM TFSI that examined the photobleach spot size-dependence of the recovered translational diffusion constant revealed the intrinsically heterogeneous nature of this RTIL. The data were consistent with a relatively small fraction of the dissociated ions partitioning into low viscosity regions of the heterogeneous system [55] and diffuse rapidly within those domains [35]. The heterogeneity referred to in reference [55] is associated with the local segregation of polar (ionic/dipolar) and nonpolar (aliphatic) regions in RTILs, while the distribution of charged species within the polar regions was found to be substantially uniform due to the ionic charges. Such a mechanism would act in concert with close-proximity ion-exchange processes in high viscosity, polar domains within the RTIL. From these bodies of work, it has emerged that there are multiple mechanisms of charge transport operating in BMIM TFSI and, almost certainly, in most other RTILs. The present work seeks to establish the dependence of these mechanisms on the RTIL constituent ion identity, and which mechanisms contribute most significantly to the conductivity. We have studied four different, structurally related RTILs (Fig. 1) and have used the chromophore RB to selectively probe the diffusional contribution to conductivity in these RTILs. We have examined these diffusional contributions as a function of temperature, which provides insights not available through single-temperature data sets.

We note that there is an important body of work that has demonstrated the utility of hole theory in addressing conductivity in ionic liquids [56–59]. Indeed, there is substantial appeal to this model and how it can be used to reconcile conductivity and viscosity, but the hole model does not take into account the intrinsic heterogeneity in RTILs or the dependence of the extra-diffusional component of conductivity on the relative density of charged moieties (*vide infra*). In addition, the sizes of the relevant charge carriers in the RTILs we examine here are such that the probability of spontaneous void spaces of the appropriate size in the RTILs under examination here is vanishingly small and is unlikely

to account for the positive deviations we observe from purely diffusive behavior [56]. In other words, while hole theory may well account for the contribution to conductivity from movement of holes, RTILs exhibit conductivity in excess of that expected from translational motion of discrete charge carriers alone. For these reasons, we believe dynamic exchange which derives from the equilibrium between free and paired ions is an important, if under-appreciated mechanism in RTILs, and present the information that follows in support of that assertion.

The diffusional contribution to molar conductivity, Λ , is typically modeled using the Nernst-Einstein (NE) equation (Eq. 1) [60]. This equation was developed to model molecular solvents and assumes a dilute homogenous system, which is known to not be the case for RTILs. Despite that limitation, the NE equation serves as a useful starting point,

$$\Lambda = \frac{z^2 N_A e^2}{k_B T} (D_T^+ + D_T^-) \quad (1)$$

where z is the ion charge, N_A is Avogadro's number, e is the charge on the electron, $k_B T$ is the thermal energy and $D_T^\pm$ are the translational diffusion constants of the ions. The translational diffusion constant for RB is determined by fitting FRAP data to the appropriate model. The FRAP measurement is the photobleaching of a circular area of the sample and measurement of the recovery of fluorescence in the photobleached region as a function of time. The resulting recovery curve is fitted using the model(s) developed by Soumpasis to determine the translational diffusion constant D_T of RB [61]. This process is detailed elsewhere [4, 62], and the three relevant models are pure diffusion (no complexation), reaction dominant (complexation dominates over diffusion) and full-reaction diffusion (both diffusion

and complexation contribute to the observed recovery). In this work, the full-reaction diffusion model is used to model the ionic liquid system and recover the quantity D_T for RB. D_T is related to system properties through the Stokes-Einstein (SE) equation (Eq. 2).

$$D_T = \frac{k_B T}{6\pi\eta r} \quad (2)$$

where $k_B T$ is the thermal energy, η is viscosity, and r is the radius of the diffusing ion [63–66]. From this information, the value of D_T for the RTIL constituent ions is scaled according to the relative hydrodynamic radii of RB and the ions (Table S1). A word is in order regarding the assumptions of this scaling process. First, the nature of the intermolecular interactions between both the RTIL cations and anions are assumed to be the same as those for the (charged) RB chromophore. This assumption is justified based on a large body of information from rotational diffusion measurements of charged chromophores in polar media. The second assumption is that the shapes of the chromophore and RTIL ions can be approximated as spherical, so that effective r values can be extracted from the hydrodynamic volumes of each species. With these caveats in mind, we are interested in using the temperature-dependence of D_T and Λ to evaluate the contributions to RTIL conductivity. The substitution of Eq. 2 into Eq. 1 yields Eqs. 3,

$$\Lambda = \frac{z^2 N_A e^2}{6\pi\eta} \left(\frac{1}{r_+} + \frac{1}{r_-} \right) \quad (3)$$

which is notable because the explicit thermal energy terms $k_B T$ cancel, leaving the viscosity, η , as the only temperature-dependent term. The temperature-dependence of RTIL viscosities has been examined extensively and this property can be modeled accurately using a Vogel-Fulcher-Tammann (VFT) equation (Eq. 4) [43, 67–69],

$$\eta(T) = \eta_0 \exp(B/(T - T_0)) \quad (4)$$

where the terms η_0 , B and T_0 are either determined empirically or based on known physical properties of the RTIL. Thus, both the magnitude and temperature-dependence of Λ can be modeled, and differences between the D_T -derived $\Lambda_{\text{diffusion}}$ and experimental $\Lambda_{\text{electrical}}$ indicate the operation of conductivity mechanism(s) other than diffusion contributing to RTIL conductivity.

We compare the temperature-dependent molar conductivity data reported in the literature to the molar conductivity data calculated from translational diffusion data measured in our laboratories (Fig. 2, Tables S2 – S4). Making these comparisons reveals that $\Lambda_{\text{diffusion}}$ alone cannot account

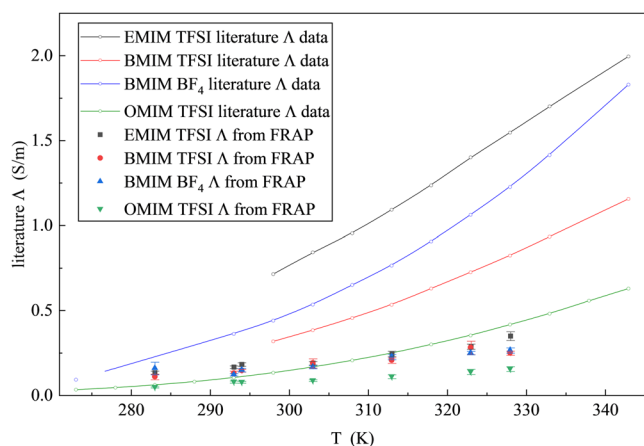


Fig. 2 Temperature-dependent conductivity data from literature data (open circles with lines) and conductivity values determined from FRAP data (solid data points with error bars) for EMIM TFSI, BMIM TFSI, BMIM BF₄, and OMIM TFSI. Color scheme shown in the legend

for $\Lambda_{\text{electrical}}$, and the temperature- and RTIL constituent identity-dependence provide insights into the non-diffusive contribution(s) to conductivity in these materials. The temperature-dependence of the experimental ionic conductivity data differs significantly from the temperature-dependence of the diffusional component, which is consistently less than the experimental conductivity.

It is instructive to consider what fraction of the electrical conductivity data is accounted for by ion diffusion. We show the ratio of diffusion-derived Λ to electrically measured Λ as a function of RTIL cation alkyl chain length in Fig. 3. For short RMIM⁺ alkyl chain lengths, diffusion plays a minority role in the observed conductivity, and for long RMIM⁺ chain lengths, diffusion becomes a much more significant contributor. We have discussed the physical origin of the non-diffusive conductivity for BMIM TFSI previously and asserted that ion exchange could account for our experimental findings. The data shown in Fig. 3 are fully consistent with that assertion; when the distance between cations and anions is small (i.e., shorter cation alkyl chains), exchange-mediated transport processes can operate efficiently, and for systems with greater spacing between ionic functionalities, exchange becomes much less efficient due to competition with geminate recombination, and the diffusional contribution to the observed electrical conductivity becomes higher.

We assert that non-diffusive conductivity is dominated by exchange, $\Lambda_{\text{electrical}} = \Lambda_{\text{diffusion}} + \Lambda_{\text{exchange}}$. Ion exchange involves dissociation and association of ionic RTIL constituents. The dissociative step in the process is an activated process, and for such a system, it is possible to determine the energy of activation from the temperature-dependence and a modified form of the Arrhenius equation.

$$\Lambda_{\text{exchange}}(T) = \Lambda_{\text{exchange}}^0 \exp(-E_a / (T - T_g)) \quad (5)$$

The exchange-mediated component to $\Lambda_{\text{electrical}}$ is indeed temperature-dependent (Fig. 4) and the activation energies, E_a , can be extracted from plots of $\ln \Lambda_{\text{exchange}}$ vs. $(T - T_g)^{-1}$. The dependence of E_a on imidazolium aliphatic chain length is consistent with the data shown in Fig. 3. In other words, the greater the distance between exchange sites, the greater the barrier to non-geminate exchange. It is important to note that we use the term $(T - T_g)^{-1}$ rather than T^{-1} in Eq. 5. The process we are concerned with is ion exchange and mobility can only proceed to an appreciable extent at temperatures above the nominal melting point. In the case of RTILs, the glass transition temperature, T_g , represents the temperature at which mobility can occur, albeit at a very limited level. Choosing T^{-1} rather than $(T - T_g)^{-1}$ would not take this thermal threshold into account, which is different for each ionic liquid. The values of E_a extracted for the RTILs studied here vary with RMIM⁺ alkyl chain length, in the range

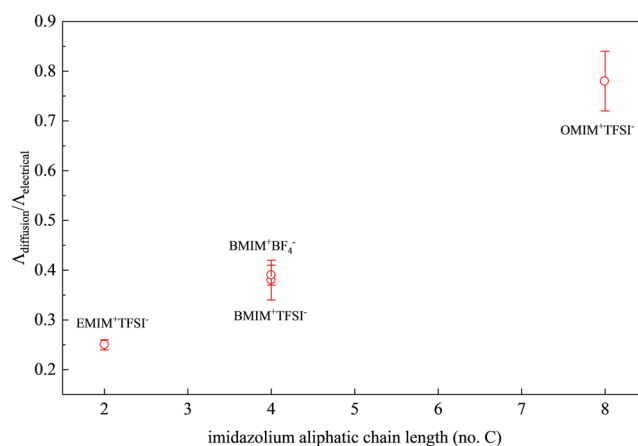


Fig. 3 Dependence of $\Lambda_{\text{diffusion}}/\Lambda_{\text{electrical}}$ ratio derived from experimental D_T data and reported Λ data to imidazolium cation aliphatic chain length at 293 K

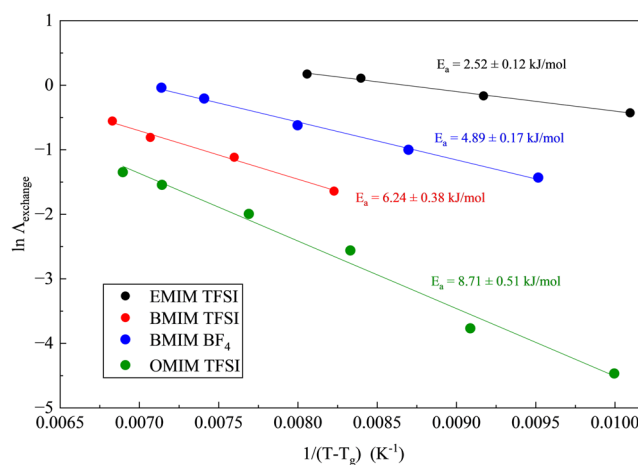


Fig. 4 Plots of $\ln \Lambda_{\text{exchange}}$ vs. $(T - T_g)^{-1}$ for EMIM TFSI (black), BMIM TFSI (red), BMIM BF₄ (blue) and OMIM TFSI (green). The activation energies associated with exchange for each RTIL is shown. Uncertainties are $\pm 1\sigma$ for the slope determined from least squares analysis

of ~ 2.5 kJ/mol to ~ 8.8 kJ/mol, increasing with inter-cation spacing. The magnitude of these values is less than a typical hydrogen bond, ~ 20 kJ/mol, again consistent with facile exchange-mediated transport. Taken collectively, these results provide strong evidence for the importance of ion exchange-mediated transport in determining the conductivity of RTILs.

Conclusions

We have reported on our experimental translational diffusion data for RB in a series of imidazolium RTILs and compared these results to corresponding published data on their electrical conductivity. This comparison, consistent with our prior work, points to the inability of conductivity in RTILs

to be accounted for solely by ionic diffusion as described in the Nernst-Einstein model. The dependence of the “excess” conductivity on RTIL cation structure and system temperature points to the important role of exchange-mediated ion transport in RTILs. Such ion exchange processes allow for facile charge transport without the need for large-scale ion diffusion. Longer cation alkyl chain length is associated with a higher diffusional contribution to the bulk molar conductivity due to the higher activation energy required for the ion exchange mechanism. The dependence of the ion exchange process on the length of the RTIL imidazolium cation alkyl chain is consistent with ion exchange-mediated charge transport, and the activation energy for the exchange process is less than a typical hydrogen bond, indicating a facile process. With these pieces of information in mind we can consider how to mediate conductivity in RTILs based on constituent ion structure and their intrinsically heterogeneous nature.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s10008-025-06512-w>.

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